

Reliability Life-Data Analysis of Device Failures (Censored Weibull)

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Reliability Engineering · Python / SciPy

Objective

Reliability engineering models how a population of devices fails over time in order to estimate lifetime and identify the failure mechanism. This analysis fits a Weibull life distribution to the time-to-failure of a device population in which most units had *not* failed when observation ended — a right-censored dataset — and extracts standard reliability metrics (characteristic life, MTTF, B10 life) plus the hazard trend that distinguishes infant-mortality from wear-out behavior.

Method

Failure times were taken from device uptime; units with no failure (Normal) were treated as right-censored at their uptime, and units with any of four failure modes (battery, CPU, network, overheating) as failures. Weibull shape (β) and scale (η) were estimated by maximum likelihood using a censored likelihood (probability density for failures, survival function for censored units). A lognormal distribution was fit the same way and compared by AIC to justify the distribution choice, and a nonparametric Kaplan–Meier estimate was computed as an assumption-free check of the fitted reliability curve.

Results

Metric	Value
Sample (failures / censored)	3000 (1190 / 1810, 60% censored)
Weibull shape β	1.85 ($\beta > 1 \rightarrow$ wear-out)
Weibull characteristic life η	181 hrs
Mean time to failure (MTTF)	161 hrs
B10 life (10% failed)	54 hrs
Distribution selection (AIC)	Weibull 15058 vs Lognormal 16016 $\rightarrow$ Weibull

Device Reliability — Censored Weibull Life-Data Analysis (1190 failures, 1810 censored)

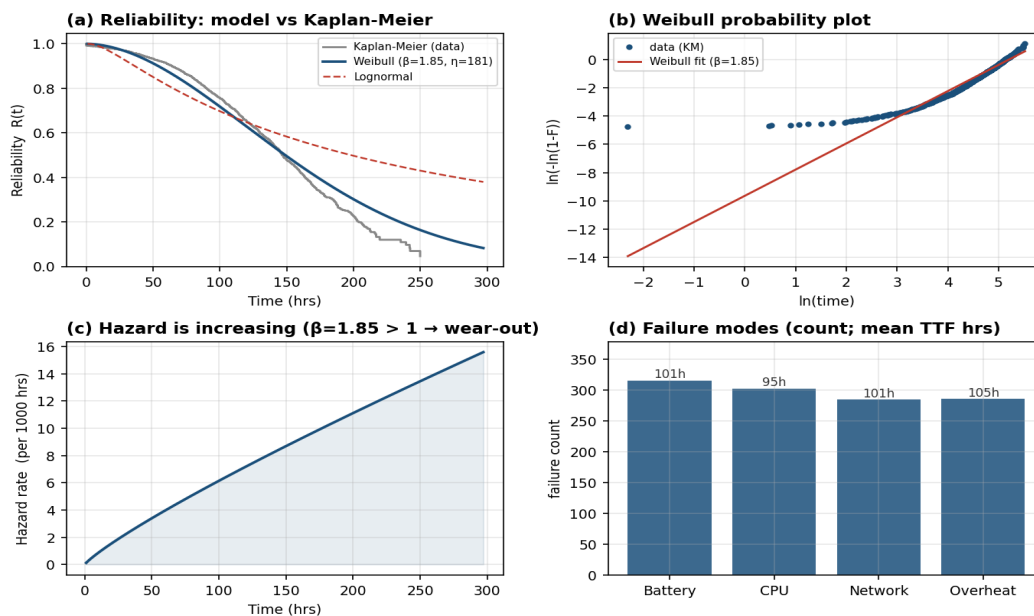


Figure 1. (a) Fitted Weibull reliability vs the Kaplan–Meier estimate and lognormal fit. (b) Weibull probability plot. (c) Increasing hazard (wear-out). (d) Failure-mode counts with mean time-to-failure.

Discussion & Conclusion

The Weibull fit gives $\beta = 1.85$ and $\eta = 181$ hrs, with an MTTF of 161 hrs and a B10 life of 54 hrs. Because β exceeds 1 the hazard rate increases with time, indicating a wear-out (aging) failure mechanism rather than early-life defects. Weibull is preferred over lognormal by AIC, and the fitted curve tracks the nonparametric Kaplan–Meier estimate closely, supporting the model. The four failure modes occur in comparable numbers with similar mean times, so no single dominant early mode is evident in this dataset; the analysis therefore emphasizes the population-level lifetime and wear-out characterization, which are the quantities used to set warranty, preventive-maintenance, and qualification targets.